KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 2 REPORT – KNOW YOUR DATA

First Impression Report

Youth Smoking and Drug Abuse Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

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1. Introduction

This report presents the Week 2 work of the project titled "Youth Smoking and Drug Abuse Analysis". Week 2 is the "Know Your Data" stage, in which the dataset is loaded and inspected before any cleaning or analysis is attempted. The purpose is to form a first impression of the dataset: its size, structure, variable types, statistical summary, missing values, duplicate records and the variety of values in each column.

For this project, participants aged 18–24 are considered the youth group. This definition is used only to focus the later analysis and does not imply that the original dataset labels all participants as youth.

2. Objective

To understand the structure, attributes and quality of the selected dataset, and to generate a First Impression report that identifies the issues to be handled during data cleaning in Week 3.

3. Activities Performed

- Loaded the dataset using Pandas and imported NumPy, Matplotlib and Seaborn.
- Examined the first and last five records using head() and tail().
- Checked the dataset shape and listed all column names.
- Identified data types and non-null counts using info().
- Generated descriptive statistics using describe().
- Checked missing values and duplicate records.
- Examined unique values in each column and the drug-consumption categories.
- Generated the First Impression report.

4. Dataset Description

The project uses the Drug Consumption dataset from the UCI Machine Learning Repository. The version used for this stage is the file drug_consumption_missing.csv, which is a structured/tabular dataset containing deliberately incomplete records so that missing-value handling can be practised in Week 3.

Attribute	Description
Dataset	Drug Consumption (UCI) – Missing Values version
Records (rows)	1,885
Columns	32
Data type	Structured / tabular
Main groups	Demographic, personality and drug-consumption variables
Youth definition	Participants aged 18–24

5. Know Your Data

5.1 Loading the Dataset

The dataset was read into a Pandas DataFrame using pd.read_csv(). The load was successful and the shape was reported as (1885, 32).

5.2 First Five Rows – First Impression

The first five records were displayed to understand the initial structure, variables and values. Demographic and personality attributes are stored as numerical values, while drug-consumption columns hold coded levels such as CL0 to CL6. The table below shows the first eight columns.

ID	Age	Gender	Education	Country	Ethnicity	Nscore	Escore
1	0.49788	0.48246	-0.05921	0.96082	0.12600	0.31287	-0.57545
2	-0.07854	-0.48246	1.98437	0.96082	-0.31685	-0.67825	1.93886
3	0.49788	-0.48246	-0.05921	0.96082	-0.31685	-0.46725	0.80523
4	-0.95197	0.48246	1.16365	0.96082	-0.31685	-0.14882	-0.80615
5	0.49788	0.48246	1.98437	0.96082	-0.31685	0.73545	-1.63340

Table 1: First five records (columns ID to Escore)

5.3 Last Five Rows

The last five records were inspected to verify that the dataset ends in the same structure as it begins. Record ID 1885 shows missing values (NaN) in Education and Nscore, giving an early sign of incomplete data.

ID	Age	Gender	Education	Country	Ethnicity	Nscore	Escore
1884	-0.95197	0.48246	-0.61113	-0.57009	-0.31685	-1.19430	1.74091
1885	-0.95197	-0.48246	NaN	-0.57009	-0.31685	NaN	1.74091
1886	-0.07854	0.48246	0.45468	-0.57009	-0.31685	1.13281	-1.37639
1887	-0.95197	0.48246	-0.61113	-0.57009	-0.31685	0.91093	-1.92173
1888	-0.95197	-0.48246	-0.61113	0.21128	-0.31685	-0.46725	2.12700

Table 2: Last five records (columns ID to Escore)

5.4 Dataset Shape

The dataset contains 1,885 rows (observations) and 32 columns (variables).

5.5 Column Names

The 32 variables are grouped as demographic attributes (Age, Gender, Education, Country, Ethnicity), personality scores (Nscore, Escore, Oscore, Ascore, Cscore, Impulsive, SS), an identifier (ID) and 19 drug-consumption variables.

#	Column	#	Column
1	ID	17	Benzos
2	Age	18	Caff
3	Gender	19	Cannabis
4	Education	20	Choc
5	Country	21	Coke
6	Ethnicity	22	Crack
7	Nscore	23	Ecstasy
8	Escore	24	Heroin
9	Oscore	25	Ketamine
10	Ascore	26	Legalh
11	Cscore	27	LSD
12	Impulsive	28	Meth
13	SS	29	Mushrooms
14	Alcohol	30	Nicotine
15	Amphet	31	Semer
16	Amyl	32	VSA

Table 3: Column names of the dataset

5.6 Dataset Information and Data Types

The info() output shows a RangeIndex of 1,885 entries with 32 columns and a memory usage of about 471.4 KB. The dataset contains 12 float64 columns, 1 int64 column and 19 object columns. Age, Gender, Education and Nscore have 1,772 non-null values, which indicates missing entries.

Column(s)	Dtype	Non-Null Count	Missing
ID	int64	1885	0
Age	float64	1772	113
Gender	float64	1772	113
Education	float64	1772	113
Country	float64	1885	0
Ethnicity	float64	1885	0
Nscore	float64	1772	113
Escore, Oscore, Ascore, Cscore	float64	1885	0
Impulsive, SS	float64	1885	0
Alcohol … VSA (19 drug columns)	object	1885	0

Table 4: Data types and non-null counts

5.7 Statistical Summary

Descriptive statistics were generated for the numerical variables. The personality scores (Nscore to SS) are standardised, with means close to 0 and standard deviations close to 1. The demographic variables Age, Gender, Education, Country and Ethnicity are stored as encoded numerical values, so their statistics describe the encoding rather than real-world quantities.

Variable	Count	Mean	Std	Min	Max
Age	1772	0.0338	0.8778	-0.9520	2.5917
Gender	1772	0.0038	0.4826	-0.4825	0.4825

Variable	Count	Mean	Std	Min	Max
Education	1772	0.0005	0.9494	-2.4359	1.9844
Country	1885	0.3555	0.7003	-0.5701	0.9608
Ethnicity	1885	-0.3096	0.1662	-1.1070	1.9073
Nscore	1772	0.0019	1.0013	-3.4644	3.2739
Escore	1885	-0.0002	0.9974	-3.2739	3.2739
Oscore	1885	-0.0005	0.9962	-3.2739	2.9016
Ascore	1885	-0.0002	0.9974	-3.4644	3.4644
Cscore	1885	-0.0004	0.9975	-3.4644	3.4644
Impulsive	1885	0.0072	0.9544	-2.5552	2.9016
SS	1885	-0.0033	0.9637	-2.0785	1.9217

Table 5: Statistical summary of numerical variables

5.8 Missing Value Check

Missing values were found in four columns only: Age, Gender, Education and Nscore. Each has 113 missing values, which is 5.99% of the records. All other columns are complete.

Column	Missing Values	Missing Percentage
Age	113	5.99%
Gender	113	5.99%
Education	113	5.99%
Nscore	113	5.99%

Table 6: Missing value summary

5.9 Duplicate Value Check

Duplicate rows were checked using df.duplicated().sum(). The result was 0, so no duplicate records are present.

5.10 Unique Value Inspection

The number of unique values per column was examined to identify categorical variables. The drug-consumption columns have 7 unique levels each (CL0 to CL6), except Semer, which has 5. Age has 6 unique values, Gender has 2 and Education has 9, confirming that these are encoded categorical quantities. ID has 1,885 unique values and acts as a record identifier.

Column	Unique Values
ID	1885
Age	6
Gender	2
Education	9
Country	7
Ethnicity	7
Nscore	49
Escore	42
Oscore	35
Ascore	41
Cscore	41
Impulsive	10

Column	Unique Values
SS	11
Semer	5
All other 18 drug columns	7 each

Table 7: Number of unique values

The categories of Alcohol, Cannabis and Coke (cocaine) were inspected in detail. Each contains all seven consumption levels CL0, CL1, CL2, CL3, CL4, CL5 and CL6, where CL0 represents never used and higher levels represent more recent or more frequent use.

6. First Impression Report

The First Impression report generated in the notebook is summarised below.

Item	Finding
Dataset name	Drug Consumption UCI – Missing Values
Number of rows	1,885
Number of columns	32
Data types	object: 19, float64: 12, int64: 1
Columns with missing values	Age, Gender, Education, Nscore (113 each)
Duplicate rows	0
Variable mix	Numerical and categorical variables
Recommendation	Data cleaning is required because missing values are present

7. Key Observations

- The dataset has 1,885 observations and 32 variables, which is a manageable size for analysis.
- It mixes numerical variables (demographic and personality scores) with categorical variables (drug-consumption levels).
- Four columns – Age, Gender, Education and Nscore – each contain 113 missing values (5.99%).
- No duplicate rows were found.
- Age, Gender, Education, Country and Ethnicity are stored as encoded numbers, which will need readable labels for visualisation.
- Drug-consumption variables use the CL0–CL6 coding and are stored as text (object) type.

8. Conclusion

The Drug Consumption dataset contains 1,885 observations and 32 variables, including numerical variables for demographic and personality attributes and categorical variables for drug-consumption levels. Initial inspection identified missing values in Age, Gender, Education and Nscore, with 113 missing values in each column, and found no duplicate rows. Therefore, the dataset requires data-cleaning procedures before detailed exploratory data analysis is performed.